Power Budget

Team Number:	Bryce Weber
Project Name:	**MEG**
Team Member Names:	**Bryce, Hattie, Rylee, Riley, Tim**
Version:	**1.0**

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or passive elements

All Major Components	Component Name	Part Number	Supply	#	Absolute	Total	Unit
	ESP32	1965-ESP32-S3-WROOM-1-N4TR-ND	+3.3V	1	500	500	mA
	Motor Driver	IFX9201SGAUMA1TR-ND	+3.3 - 36V	1	13	13	mA
	Motor	2183-3042-ND	+12V	1	750	750	mA

B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the design.

+12V Power Rail	Component Name	Part Number	Supply	#	Absolute	Total	Unit
	Motor Driver	IFX9201SGAUMA1TR-ND	+3.3 - 36V	1	13	13	mA
	Motor	2183-3042-ND	+12V	1	750	750	mA
						0	mA
						0	mA
						0	mA
					Subtotal	763	mA
					Safety Margin	25%	
					Total Current Required on +12V Rail	953.75	mA
c1. Regulator or Source Choice	Plug-in Wall Supply	364-1276-ND	90 ~ 264 VAC	+12V	1000	1000	mA
					Total Remaining Current Available on +12V Rail	46.25	mA

+3.3V Power Rail	Component Name	Part Number	Supply	#	Absolute	Total	Unit
	ESP32	1965-ESP32-S3-WROOM-1-N4TR-ND	+3.3V	1	500	500	mA
	Motor Driver	IFX9201SGAUMA1TR-ND	+3.3 - 36V	1	13	13	mA
						0	mA
						0	mA
					Subtotal	513	mA
					Safety Margin	25%	
					Total Current Required on +3.3V Rail	641.25	mA
c4. Regulator or Source Choice	+3.3V low-dropout regulator	LM2575D2T-3.3R4GOSTR-ND	+3.3V	1	1000	1000	mA
					Total Remaining Current Available on 3.3V Rail	358.75	mA

C. For each power rail above, select a specific voltage regulator using the same process as for major component selection. Confirm that the Total Remaining Current Available on

D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the regulators for all of the power rails simultaneously. If

External Power Source 1	Component Name	Part Number	Supply	Output	Absolute	Total	Unit
Power Source 1 Selection	Plug-in Wall Supply	364-1276-ND	90 ~ 264 VAC	+12V	2000	2000	mA
Power Rails Connected to External Power Source 1					1000	0	mA
						0	mA

External Power Source 1	+3.3V low-dropout regulator	LM2575D2T-3.3R4GOSTR-ND	+5V - 20V	1	1000	1000	mA
	Total Remaining Current Available on External Power Source 1					1000	mA
External Power Source 2	**Component Name**	**Part Number**	**Supply**	**Output**	**Absolute**	**Total**	**Unit**
Power Source 2 Selection	Battery	(full part number)	+9V	-9V	500	500	mA
Power Rails Connected to External Power Source 2	-5V Regulator	(full part number)	(range)	1	500	500	mA
	Total Remaining Current Available on External Power Source 2					0	mA
E. Calculate Battery Life (if applicable). For each battery, also check the worst-case lifetime of the battery by indicating the capacity in mAh.							
	Component Name	**Part Number**	**Supply**		**Capacity**	**Required**	
	Battery	(full part number)	+12V		500	500	
					Battery Life	1	hours
Notes							
External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).							
If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit							